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Paleomagnetism of the Late Cambrian *Crepicephalus-Aphelaspis* trilobite zone boundary in North America—divergent poles from isochronous strata

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Paleomagnetic samples from the Nolichucky Formation (Late Cambrian), sampled at two sites in the Valley and Ridge Province of east Tennessee, yield a possibly penecontemporaneous characteristic magnetization that appears to reside in detrital magnetite. The paleomagnetic pole positions are "Paleozoic", but differ: site I, lat. 41°N, long. 109°E, $dp = 1°$, $dm = 2°$; site II, lat. 39°N, long. 131°E, $dp = 4°$, $dm = 7°$. The difference in poles reflects a significant difference in declination between the site-mean directions, and this declination difference probably reflects relative tectonic rotation as the sites are in different thrust sheets. The paleontologic age of both sections is exceptionally well-constrained as they are sampled across an abrupt "biomere boundary" between contrasting trilobite faunas. Comparison of these results with paleomagnetic data from coeval strata elsewhere in North America reveals gross discrepancies, so that at least some of the published data must reflect remagnetization and/or tectonic rotation.

1. Introduction

Paleomagnetic poles from the Cambrian of North America are scattered, and despite a recent flurry of papers (e.g. [1–3]) a unique solution to the Cambrian apparent polar wander (APW) path does not seem to be emerging. This scatter has been variously attributed to rapid APW [2,3], remagnetization [4], tectonic rotation [1], or non-dipole behavior of the geomagnetic field [5]. As Roy and Anderson [6] point out, tests between such hypotheses must involve paleomagnetic data from *contemporaneous* formations at numerous, widely-spaced geographic localities.

Palmer [7,8] has defined a "biomere" as a regional biostratigraphic unit that is "bounded by abrupt, non-evolutionary changes in the dominant elements of a single phylum", and he recognizes several such biomeres in the Cambrian of North America (Fig. 1). Paleontologically the significance of biomeres is controversial [8]; for the purposes of paleomagnetism their utility stems from the abruptness of their boundaries. The change in faunas can be pinpointed within a meter stratigraphically, yielding a degree of relative time control between different sections that is normally impossible to achieve with biostratigraphy. Palmer [8] estimates that the transition between the *Crepicephalus* and *Aphelaspis* faunas (Fig. 1) took less than 6000 years; even if this estimate is wrong by an order of magnitude, the duration of this transition is still very short on the scale of APW. In the spirit of Roy and Anderson's [6] suggestion, paleomagnetic data from near the *Crepicephalus-Aphelaspis* faunal boundary will be compared for various localities throughout North America.

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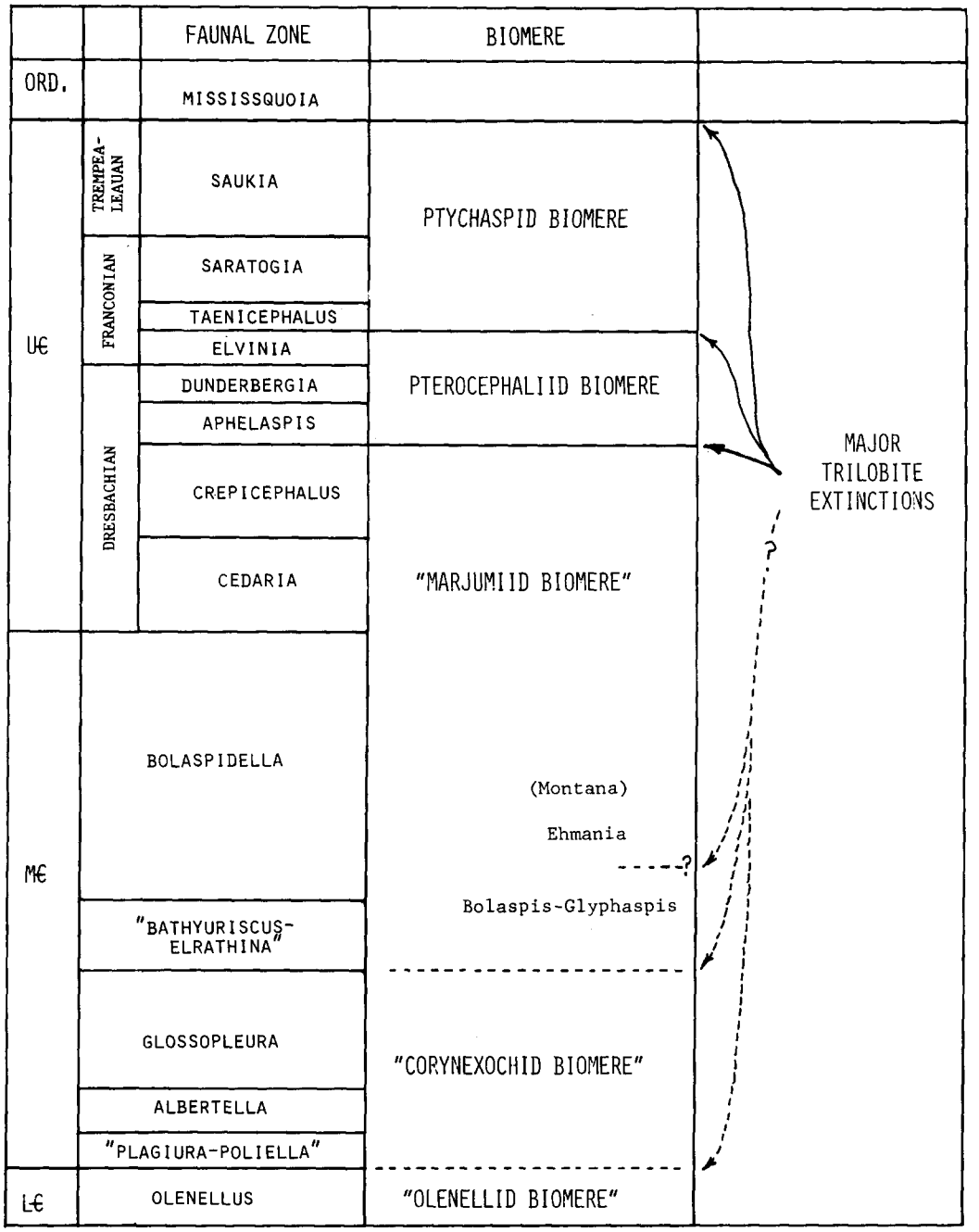


Fig. 1. Stages and faunal zones in the Cambrian of North America, showing locations of biomere boundaries (from [8] and A.R. Palmer, personal communication, 1981).

2. New results

Data from two new sections sampled across this biomere boundary in eastern Tennessee are presented here; the general area is Palmer's [8] location 8. The sampling area is within the Valley and Ridge structural province, which is here characterized by imbricate thrust sheets that verge

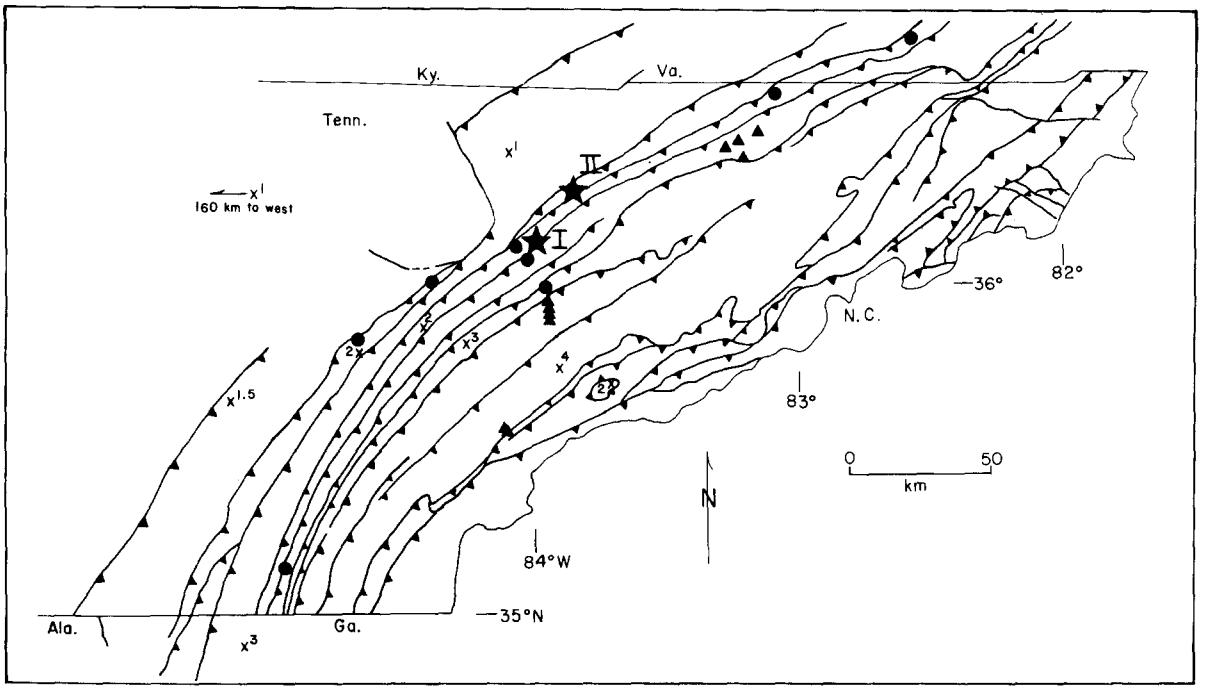


Fig. 2. General tectonic and index map showing sample localities in the Valley and Ridge Province of eastern Tennessee. Mapping of the thrust faults is generalized from Rodgers [21]. Solid stars are the sites where the Late Cambrian Nolichucky Formation was sampled (this study); site numbers correspond to Table 1. Also shown for comparison are the sites in Ordovician strata sampled by Watts and Van der Voo ([24]; solid triangles), and the sites in the Cambrian Rome Formation sampled by Watts et al. ([2]; solid circles). In addition, sites where conodont alteration indices (CAI's) have been reported from Ordovician strata are shown by ×'s [19]; the adjacent number is the alteration index. Higher alteration indices correspond to greater heating, probably from deeper burial.

westward (Fig. 2). The thrusting is rooted in Early Cambrian or Late Precambrian shales and siltstones and is Alleghenian (late Paleozoic) in age [9]. Both sections are in limestone beds in the Late Cambrian Nolichucky Formation; one (site I) is in a spectacular road cut on I-75 about 16 km north of Knoxville, and the other (site II) is along a shoreline exposure on Norris Lake (Fig. 2). The sites are in different thrust sheets.

Both sections consist of gray limestone with some interbedded green shale. The limestones are heterogeneous, mostly grainstones, and probably of shallow-water origin; they include all gradations of oolite, intraformational conglomerate and trilobite debris. Patchy replacement of calcite by ferroan dolomite ranges from present to pervasive. Stylolites are fairly common, and range from microstylolites at grain contacts to large, throughgoing features generally parallel to bedding. They tend to be localized at contrasts in lithologic type. Pyrite is common, occurring either as disseminated blebs or as euhedral cubes ~ 1 mm on a side. Commonly, pyrite blebs are concentrated along stylolites. Rare glauconite occurs.

Oxidation from present weathering is locally common in these rocks. At site II, the lakeshore exposure, oxidation is omnipresent, but the rocks at site I (in a deep road cut) are much less altered in general. Where present, the oxidation generally imparts an orange cast to the rock from the formation of limonite. Oxidation tends to become most intense at the surface of a sample, but it also commonly follows stylolites and bedding contacts. The pyrite and ferroan dolomite are most readily oxidized; pyrite is commonly altered to pseudomorphs of dark red hematite, and intermediate stages can be found where a core of unaltered pyrite is preserved within a hematite rim. The

ferroan dolomite in early stages of oxidation acquires an orange cast. In extreme cases it is replaced by a calcite pseudomorph ("dedolomite"); on etched surfaces, these appear as rhombic molds outlined by iron oxides.

Oriented cores 2.5 cm in diameter were collected with a portable gasoline-powered drill similar to that described by Doell and Cox [10], and as many as possible (1 or 2) specimens ~2.5 cm long were trimmed from each sample. Trimmed specimens were rinsed briefly with dilute HCl before measuring, to minimize possible contamination from the drilling and sawing. Samples were collected across the biomere boundary in stratigraphic sequence at ~0.3- to 1-m intervals. Magnetic measurements were made on an ScT 2-axis cryogenic magnetometer interfaced to a computer, allowing on-line data reduction, as described by Gillett et al. [11].

Initially, one specimen from each sample was subjected to progressive thermal demagnetization at 6 steps between 200 and 450°C or until the specimen became too weak to measure; a step at 500°C was also performed on the specimens from site II. A steep component of magnetization, probably resulting from the present field, is removed by about 250°C; thereafter, a shallow, positive, southeasterly component is revealed which decays to the origin upon further demagnetization (Fig. 3a). At both sites, the initial step at 200°C causes about 50% intensity loss; at site I, the intensity remaining at 450°C is a few percent of the NRM, whereas at site II about 10% of the NRM intensity is left. The larger residual intensity at site II may be due to a component in the recently formed hematite in that section. The behavior of these samples at the high demagnetization steps is erratic, however, and no consistent component can be identified on the vector diagrams.

Alternating-field (AF) demagnetization was performed on a few duplicate specimens from both sites. The specimens were first demagnetized at 200°C to destroy any present-field component that might reside in goethite from weathering, and then subjected to steps of 5, 10, 20, 35, and 50 mT (1 mT = 10 Oe). In both sections, the loss of intensity was fairly small to 10 mT, and then accelerated so that less than 50% of the intensity at 200°C remained after demagnetization to 50 mT. A component similar to the one seen upon thermal demagnetization decayed toward the origin upon AF demagnetization (Fig. 3b).

Using the cryogenic magnetometer as described by Gillett et al. [11], bulk susceptibility was mea-

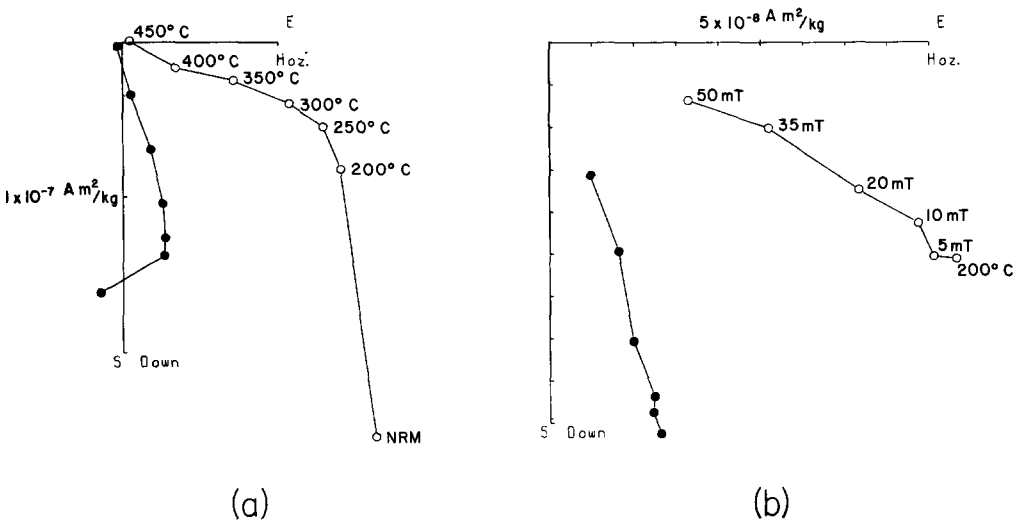


Fig. 3. Vector demagnetization diagrams [12] of gray limestone from the Late Cambrian Nolichucky Formation at site I. Closed (open) circles represent declination (inclination), respectively; 1 A $m^2/kg = 1$ emu/g. (a) Thermal demagnetization; (b) AF-following-thermal demagnetization; note that the NRM is omitted.

sured at NRM and 200°C on these duplicate specimens, and on the corresponding original specimens that had been demagnetized at 450 or 500°C. Although the scatter in the data is large, there is no significant change in susceptibility to 200°C, but it then increases by a factor of two to three at the high demagnetization steps. Probably this behavior results from the formation of some magnetite during heating, perhaps from oxidation of the pyrite. Another possibility is reduction of some of the recently formed hematite or goethite [13,14].

The demagnetization data are interpreted as follows. A characteristic component [15] with low positive inclination is revealed by the thermal demagnetization; the moderate unblocking temperature of this component, coupled with its response to AF demagnetization, suggests that it resides in magnetite. This magnetite is probably detrital; it could not form authigenically as the rocks are too sulfur-rich, making pyrite the stable phase (e.g. [16,17]). Also on chemical grounds, hematite is not likely to be important except as a product of present weathering.

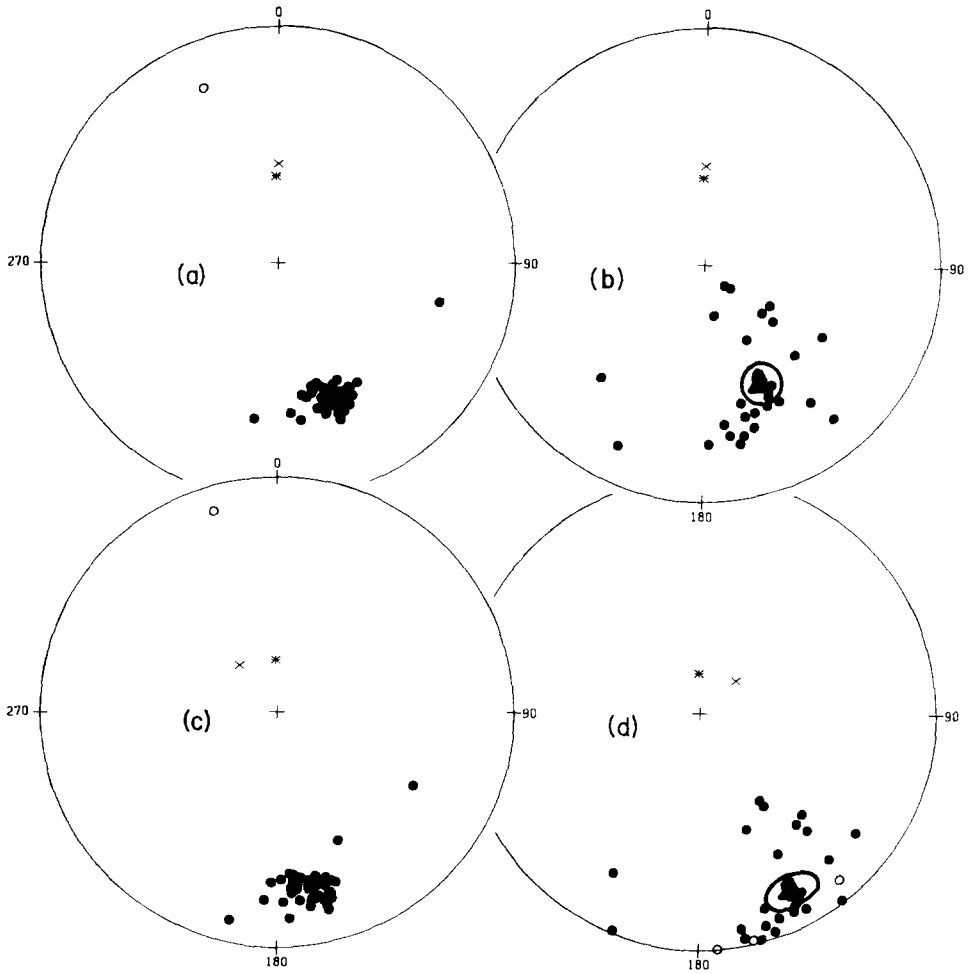


Fig. 4. Equal-area plot of characteristic directions of magnetizations, found by least-squares fits [11] from the Nolichucky Formation. (a) geographic coordinates, site I; (b) geographic coordinates, site II; (c) stratigraphic (dip-corrected) coordinates, site I; (d) stratigraphic coordinates, site II. Stars in (b) and (d) are site means; (separate site-mean stars in (a) and (c) would not be visible).. * (asterisk) is position of present axial dipole (present local) field.

TABLE 1

Summary of paleomagnetic results from calcarenites in the Upper Cambrian Nolichucky Formation, eastern Tennessee

Section	Average attitude (strike, dip)	Geographic coordinates [a]					Stratigraphic coordinates [b]					North pole (lat., long.)	Error oval (dp, dm) [g]
		D [c]	I [c]	k [d]	α_{95} [e]	N/N_{tot} [f]	D [c]	I [c]	k [d]	α_{95} [e]	N/N_{tot}		
Site I	127°, 24°W	159°	38°	133	2°	46/48	170°	24°	99	2°	46/48	41°N, 109°E	1°, 2°
		($k_1=-110$, $k_2=-51$) [h]					($k_1=-67$, $k_2=-43$)						
		(b =7.2 *, $g=1147$) [i]					(b =2.3, $g=953$)						
Site II	62°, 27°E	154°	44°	16	7°	28/30	153°	17°	16	7°	28/30	39°N, 131°E	4°, 7°
		($k_1=-20$, $k_2=-6.5$)					($k_1=-21$, $k_2=-6.5$)						
		(b =12.1 *, $g=72$)					($b=12.2$ *, $g=73$)						
						mode:	157°	8°	—	5°	20/30	44°N, 129°E	3°, 5°

[a] Geographic coordinates are not corrected for bedding dip.

[b] Stratigraphic coordinates are corrected for bedding dip.

[c] D = declination; I = inclination.

[d] Unbiased estimate of concentration parameter [29]; k = is not given where modes were calculated.

[e] Where Fisher [29] statistics calculated, half-angle of 95% confidence cone. Where modes calculated, estimate of half-angle of 95% confidence ("β_{95}") suggested by Van Alstine [18].

[f] N_{tot}, total number of samples. For Fisher statistics, N is the number of samples included in the mean. For modal analysis, N is the number of samples with a Fisherian mean equal to the mode; this mean is used to calculate β_{95} [18].

[g] dp = semiminor axis of error oval; dm = semimajor axis of error oval.

[h] k_1, k_2 = Bingham [30] concentration parameters, using notation of Onstott [31].

[i] Test statistics for circular symmetry based on Bingham parameters [32, pp. 277–378]. Modes [18] were calculated where these statistics indicated that there was a significant departure from circular symmetry (see Gillett et al. [11]).

* Indicates a significant departure from a circularly symmetric axial distribution at the 99% confidence level.

Following the method described by Gillett et al. [11], least-squares fits were made to visually-identified linear segments of the demagnetization paths to define the characteristic direction (Fig. 4; Table 1). One sample from site I is nearly antipodal to the others; whether this reflects a true polarity reversal is not clear at this point. If so, the polarity zone is very thin as the stratigraphic intervals are ~ 1 m. More likely, the sample is misoriented but the demagnetization path is equivocal in this respect. For now, the sample is excluded from the mean.

The additional scatter apparent in site II probably results from its more intense weathering. After discarding two obvious outliers the resulting distribution is significantly non-circular (Table 1); hence a mode [18] was calculated for this distribution. The mode does not differ much from the mean in declination, but it has a shallower inclination (Table 1). Although this results in the mode being farther from the present-field direction than the mean, the modal inclination also differs from the inclination found in the well-grouped samples at site I. Hence, the mean is preferred as an estimate at this time. In any case, the declinations of the two sites are statistically different.

3. Discussion

The characteristic magnetization found in the Nolichucky Formation has a direction similar to a reversed-polarity Paleozoic direction. The conodont color alteration index (CAI) for Ordovician strata overlying the two sites is less than 2 (Fig. 2), indicating that burial heating has probably not exceeded 100°C over 10^8 years [19]. Such low-level heating most probably could not cause remagnetization, as the observed unblocking temperatures are >400°C [20]. Hence, it seems likely that the characteristic magnetization is a penecontemporaneous detrital remanent magnetization (DRM) residing in magnetite.

Although the mean directions of magnetization from sites I and II agree somewhat better before structural correction, suggesting that the characteristic magnetization might be post-thrusting, this is probably fortuitous as outlined above. However, if the characteristic magnetization does reflect a viscous partial thermoremanent magnetization (VPTRM [20]) imposed, say, during uplift at the end of the Paleozoic, such remagnetization must be extremely easy to accomplish. Where Gillett [4] inferred a magnetite VPTRM in Paleozoic limestones in southern Nevada, the conodont CAI's are about 4.5, indicating a minimum of ~250°C sustained for at least 10^8 years. Therefore, if the Nolichucky results reflect remagnetization, complete remagnetization of ancient strata can be expected to be pervasive, and the conclusions of this paper regarding the inadequacy of the present paleomagnetic data base in Early Paleozoic strata become even more forceful.

The statistically significant difference in declination between the two sites, after correction for bedding, suggests that relative rotation about a vertical axis has occurred. As the sites are in different thrust sheets, the declination discrepancy would suggest that relative rotation has occurred between the thrust sheets. This seems reasonable; even if the entire thrust sheet does not rotate as a rigid block, internal distortion and fracturing would be expected to occur within it, and these might lead to local rotations. Indeed, internal folding is ubiquitous within the thrust sheets [21,22].

Another, purely geometric phenomenon may lead to apparent vertical-axis rotations, however. The conventional structural correction, in which bedding is rotated back to horizontal about a horizontal axis (strike line), yields incorrect results when the strata have been subjected to a net rotation about an inclined axis (e.g. [23]). The effect of the conventional correction is to introduce a spurious anomaly in declination: an "apparent vertical-axis rotation". As inclination with respect to bedding is always fixed, this has the effect of introducing additional scatter into declination. True vertical-axis rotations are probably also common in orogenic belts, however, and as MacDonald [23] notes, distinguishing true from apparent rotations may be very difficult. It is possible that the declination differences observed between the Nolichucky sites may have resulted from such geometric artifacts, as the attitudes at the two sites are very different (Table 1). Whether true or apparent, however, the declination dif-

ference introduces ambiguity into the interpretation of these directions.

3.1. *Previous work on Early Paleozoic rocks, east Tennessee*

In this part of the Valley and Ridge Province, Watts and Van der Voo [24] reported paleomagnetic data from Ordovician strata, and Watts et al. [2] presented data from the Middle Cambrian Rome Formation (Table 2a and Fig. 5a). These strata were shown to have stable, pre-thrusting characteristic magnetizations that reside in hematite. These authors concluded that little if any vertical-axis rotation due to thrusting had taken place. In the Rome Formation, Watts et al. [2] presented data from sites in a number of thrust sheets and addressed two kinds of hypothetical vertical-axis rotation; one was a consistent, large-scale oroclinal bending of the Appalachians in this area, and the other was relative rotations between the thrust sheets. They appear to have conclusively ruled out a rotation of the former type, and they thought the latter type unlikely, based in part on geometric arguments concerning the freedom to rotate of a rigid thrust sheet. They did not address the problem of apparent rotations. The data presented in this paper, however, show that significant differences in declination between sites occur locally. Further, examination of the site-mean data presented by Watts et al. [2] shows much scatter in declination (e.g., their figs. 5 and 6). These differences are not statistically significant because of the small number of samples per site; however, the discrepancies probably merit further investigation. Geometric arguments based on rigid thrust sheets may be misleading.

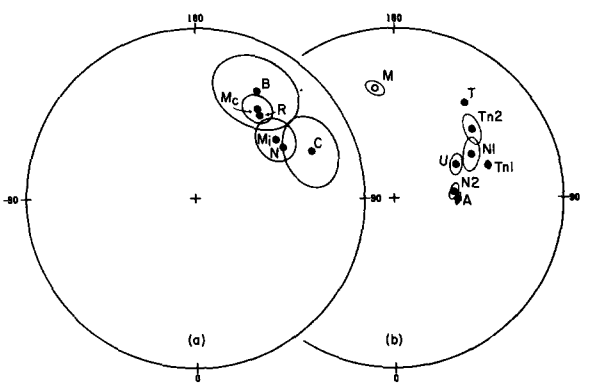


Fig. 5. Equal-area plots of paleomagnetic poles, centered on north geographic pole. Solid circles on northern (upper) hemisphere, open on southern hemisphere. (a) Paleomagnetic pole positions from Early Paleozoic strata in the Valley and Ridge Province of eastern Tennessee. The data and abbreviations are listed in Table 2a, and site locations in Fig. 2. (b) Pole positions from at or near the *Crepicephalus-Aphelaspis* trilobite zone boundary (Dresbachian; Late Cambrian) from throughout North America. Data and abbreviations are listed in Table 2b.

Stratigraphically, the Nolichucky lies between the Rome and overlying Ordovician strata, and it is interesting to compare its mean direction with those from the other rock units (Fig. 5a and Table 2a). Mean directions from all formations have been estimated by averaging the site-mean poles, which will tend to average out relative rotations if the rotations have been random. The pole from the Middle Cambrian Rome Formation agrees very well with those from the Late Ordovician Moccasin and Bays Formations. However, the average pole from the Late Cambrian Nolichucky derived herein differs from those strata, but it agrees reasonably well with the pole from the mid-Ordovician Chapman Ridge Formation. Watts and Van der Voo [24] tentatively interpreted the difference between the Chapman Ridge and the younger Ordovician units as reflecting apparent polar wander, but they noted that, as all the Ordovician magnetizations resided in hematite, later remagnetization was also possible. At face value, the Nolichucky results support their hypothesis of a "loop" of the APW path between Rome and Moccasin-Bays time, especially as the magnetization of the Nolichucky apparently resides in magnetite. However, in view of the declination discrepancies demonstrated above, I hesitate to interpret these directions in terms of polar wander, and this hesitation is strengthened by discrepancies, discussed below, with Late Cambrian poles from the rest of North America. Paleomagnetic data from Ordovician rocks in additional thrust sheets would be very desirable in determining whether vertical-axis rotations have contributed to the differences between the poles.

TABLE 2

Comparison of paleomagnetic data

Rock unit	Location	North pole (lat., long.)	Error (*dp*, *dm*)	Reference	Remarks
Nolichucky Fm., site I (Tn1)	east Tennessee	41°N, 109°E	1°, 2°	this study	
Nolichucky Fm., site II (Tn2)	east Tennessee	39°N, 131°E	4°, 7°	this study	mean instead of mode; see text
Nolichucky Fm. (N)	east Tennessee	40°N, 120°E	–	this study	mean of sites I and II
(*a*) *Data reported from other Early Paleozoic strata in the Valley and Ridge Province of eastern Tennessee*					
Rome Fm. (R)	east Tennessee	38°N, 142°E	–	[2]	Cambrian; mean of all poles
Chapman Ridge Fm. (C)	east Tennessee	27°N, 112°E	$\alpha_{95} = 15°$	[23]	Ordovician; mean of poles
Moccasin Fm. (M_c)	east Tennessee	36°N, 145°E	$\alpha_{95} = 7°$	[23]	high blocking-temperature component; Ordovician
Moccasin Fm. (M_i)	east Tennessee	41°N, 126°E	$\alpha_{95} = 10°$	[23]	intermediate blocking-temperature component
Bays Fm. (B)	east Tennessee	28°N, 150°E	$\alpha_{95} = 19°$	[23]	Ordovician; mean of poles
(*b*) *Data from North America from at or near the Dresbachian* (*Late Cambrian*) Crepicephalus-Aphelaspis *trilobite zone boundary* (*see text*)					
Bonanza King Fm., section Q (N1)	south Nevada	47°N, 119°E	4°, 8°	[4]	other evidence suggests these sections have been remagnetized and have also undergone relative tectonic rotation
Bonanza King Fm., section U (N2)	south Nevada	61°N, 96°E	2°, 4°	[4]	
Orr Fm. (U)	west Utah	56°N, 118°E	3°, 5°	[11]	possibly remagnetized
Cap Mtn–Lion Mtn. Fms. (T)	Texas	31°N, 143°E	–	[3]	mean of 6 site means
Bonneterre Dolostone (M)	Missouri	35°S, 170°W	3°, 5°	[24]	
Abrigo Fm. (A)	Arizona	59°N, 89°E	1°, 3°	[25]	

* Abbreviations after rock units correspond to points in Fig. 5.

3.2. Previous results from coeval rocks

A number of authors have reported data from at or near this biomere boundary throughout North America [3,4,11,25,26]. As has been emphasized, the paleontologic data very tightly constrain the relative ages of these strata; hence, the scatter of paleomagnetic poles from nearly coeval rocks (Fig. 5b and Table 2b) demonstrates that some, or all, of these strata do not record the Cambrian geomagnetic field. Possible explanations for the discrepancies will be advanced as these results are discussed in turn.

Gillett [4] reported paleomagnetic data from miogeoclinal strata athwart this biomere boundary in southern Nevada; it was concluded on other evidence that those rocks had been remagnetized. Other data, from western Utah, from this biomere boundary were reported by Gillett et al. [11], where it was also suggested that the rocks might be remagnetized. In view of the suspected remagnetization of the sections from western North America, it is ironic that they yield pole positions similar to those from the Nolichucky (Fig. 5b).

Watts et al. [3] in a study of cratonic Late Cambrian strata from the Llano uplift in Texas, also crossed this biomere boundary; the characteristic magnetizations from these strata (Cap Mountain–Lion Mountain Members) yield a pole near the pole from site II (Fig. 5b); at face value, this would suggest that this site is the less rotated. The magnetizations from the Llano reside in hematite, however, and are from a thin, cratonic section that contains many major unconfirmities. Thus a diagenetic chemical remanent remagnetization of those strata cannot be ruled out at this point.

Beales et al. [25] reported paleomagnetic data from the Bonneterre Dolostone of the Ozarks; a *Crepicephalus* fauna has been collected from the Bonneterre [27] so that it is very near the age of the biomere boundary. The directions from the Bonneterre are similar to those from the underlying Lamotte Sandstone [28]; they are very aberrant, however, (Fig. 5b), when compared with the other Cambrian results. This discrepancy merits further study as the Bonneterre-Lamotte pole is very far from other reported Paleozoic paleomagnetic poles. The magnetometer used in the study of the Bonneterre, however, was near its limits of detection. Further, the Lamotte direction is based upon AF demagnetization only although the magnetization in that formation apparently resides in hematite. It is possible, therefore, that the observed direction of magnetization reflects a vector sum of unresolved components.

Paleomagnetic data were reported from the Abrigo Formation of southern Arizona by Elston and Bressler [26]. These authors quoted an approximate "Dresbachian" (early Late Cambrian; Fig. 1) age for the Abrigo, so that it is near in age to the biomere boundary. The magnetization resides in hematite; samples were collected from a "hematitic oolite" about 30 cm thick. The direction is also discrepant, but in the opposite sense from the Bonneterre. It seems likely that this magnetization could reflect relatively late remagnetization. Hematite is not a likely primary component of an oolite, as carbonates tend to be reduced; however, hematization commonly results from secondary oxidation of pyrite or ferroan dolomite. In other limestones from the great Basin, for example, the author has commonly observed diagenetic replacement of ooids by aggregates of large, euhedral rhombs of ferroan dolomite or ankerite. These rhombs are very susceptible to iron oxidation if conditions become oxidizing; for example, present-day outcrops are generally pervasively oxidized from weathering. Petrographic study to determine the relative timing of the hematite vis a vis the diagenetic history of the Abrigo is obviously desirable. Further, as the Abrigo is in the southern Basin and Range province it may also have undergone vertical-axis rotations.

4. Conclusions

In the Valley and Ridge Province of eastern Tennessee, Late Cambrian strata of the Nolichucky Formation have a probably penecontemporaneous characteristic magnetization, residing in magnetite, that shows apparent relative rotation around a vertical axis to have occurred between different thrust sheets. This rotation may be a geometric artifact, but in any case it introduces ambiguity into the interpretation of directions. These results

demonstrate that paleomagnetic data from thrust belts must be treated with great discretion; sites should be sampled in several thrust sheets, with good stratigraphic control between the sites, and enough samples should be taken at each site to insure that any significant rotations could be identified.

Paleomagnetic poles derived from Late Cambrian strata that are nearly coeval with the *Crepicephalus-Aphelaspis* trilobite zone boundary from throughout North America have a very scattered distribution, and apparently few if any record a penecontemporaneous magnetization. In some cases, remagnetization or tectonic rotations seem very likely, and in other instances these explanations have not been ruled out; no data can confidently be regarded as indicative of a Cambrian paleomagnetic pole. Before rapid APW or anomalous behavior of the geomagnetic field is invoked to explain aberrant paleomagnetic data, the more prosaic explanations of remagnetization or tectonic rotation must be rigidly excluded.

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References

1 S.L. Gillett and D.R. Van Alstine, Paleomagnetism of Lower and Middle Cambrian sedimentary rocks from the Desert Range, Nevada, J. Geophys. Res. 81 (1979) 4475–4489.
2 D.R. Watts, R. Van der Voo and R.B. French, Paleomagnetic investigation of the Cambrian Waynesboro and Rome Formations of the Valley and Ridge Province of the Appalachian Mountains, J. Geophys. Res. 85 (1980) 5331–5343.
3 D.R. Watts, R. Van der Voo and S.C. Reeve, Cambrian paleomagnetism of the Llano Uplift, Texas, J. Geophys. Res. 85 (1980) 5316–5330.
4 S.L. Gillett, Magnetization and remagnetization processes in some Early Paleozoic limestones from the Great Basin, Ph.D. Thesis, State University of New York at Stony Brook, N.Y. (1981).
5 D.R. Van Alstine, Apparent polar wandering with respect to North America since the late Precambrian, Ph.D. Thesis, California Institute of Technology, Pasadena, Calif. (1979).
6 J.L. Roy and P. Anderson, An investigation of the remanence characteristics of three sedimentary units of the Silurian Mascarene Group of New Brunswick, Canada, J. Geophys. Res. 86 (1981) 6351–6368.
7 A.R. Palmer, Biomere—a new kind of biostratigraphic unit, J. Paleontol. 39 (1965) 149–153.
8 A.R. Palmer, Biomere boundaries re-examined, Alcheringa 3 (1979) 33–41.
9 R. Van der Voo, Age of the Alleghenian folding in the central Appalachians, Geology 7 (1979) 297–298.
10 R.R. Doell and A. Cox, Palaeomagnetic sampling with a portable coring drill, in: Methods in Palaeomagnetism, D.W. Collinson, K.M. Creer and S.K. Runcorn, eds. (Elsevier, Amsterdam, 1967) 21–25.
11 S.L. Gillett, V.C. Hover and J.J. Papike, The Notch Peak contact metamorphic aureole, Utah, I. Paleomagnetism of the metasedimentary rocks and the quartz monzonite stock, J. Geophys. Res. (in press).
12 J.L. Roy and J.K. Park, The magnetization process of certain red beds: vector analysis of chemical and thermal results, Can. J. Earth Sci. 11 (1974) 437–471.
13 P.N. Shive and J.F. Diehl, Reduction of hematite to magnetite under natural and laboratory conditions, J. Geomagn. Geoelectr. 29 (1977) 345–354.
14 P.N. Shive and J.F. Diehl, Thermomagnetic analysis of natural and synthetic hematite, Geophys. Res. Lett. 4 (1977) 159–162.
15 J.D.A. Zijderveld, A.C. demagnetization of rocks: analysis of results, in: Methods in Palaeomagnetism, D.W. Collinson, K.M. Creer and S.K. Runcorn, eds. (Elsevier, Amsterdam, 1967) 254–286.
16 R.A. Berner, Principles of Chemical Sedimentology, (McGraw-Hill, New York, N.Y., 1971) 240 pp.
17 R.A. Berner, A new geochemical classification of sedimentary environments, J. Sediment. Petrol. 51 (1981) 359–365.
18 D.R. Van Alstine, Analysis of the modes of directional data with particular reference to palaeomagnetism, Geophys. J.R. Astron. Soc. 61 (1980) 101–113.
19 A.G. Epstein, J.B. Epstein and L.D. Harris, Conodont color alteration—an index to organic metamorphism, U.S. Geol. Surv. Prof. Paper 995 (1977).
20 G. Pullaiah, E. Irving, K.L. Buchan and D.J. Dunlop, Magnetization changes caused by burial and uplift, Earth Planet. Sci. Lett. 28 (1975) 133–143.
21 J. Rodgers, Geologic map of east Tennessee with explanatory text, Tenn. Div. Geol., Bull. 58 (1953) 168 pp.

22 G.D. Swingle, Structural geology of Knox County, Tennessee, Geology of Knox County, Tenn. Div. Geol., Bull. 70 (1973) 63–73.
23 W.D. MacDonald, Net tectonic rotation, apparent tectonic rotation, and the structural tilt correction in paleomagnetic studies, J. Geophys. Res. 85 (1980) 3659–3669.
24 D.R. Watts and R. Van der Voo, Paleomagnetic results from the Ordovician Moccasin, Bays, and Chapman Ridge Formations of the Valley and Ridge Province, eastern Tennessee, J. Geophys. Res. 84 (1979) 645–655.
25 F.W. Beales, J.C. Carracedo and D.W. Strangway, Paleomagnetism and the origin of Mississippi Valley-type ore deposits, Can. J. Earth Sci. 11 (1974) 211–223.
26 D.P. Elston and S.L. Bressler, Paleomagnetic poles and polarity zonation from Cambrian and Devonian strata of Arizona, Earth Planet. Sci. Lett. 36 (1977) 423–433.
27 C. Lochman-Balk, The Cambrian of the craton of the United States, in: Cambrian of the New World, C.H. Holland, ed. (Wiley Interscience, New York, N.Y., 1971) 79–167.
28 S.A. Al-Khafaji and S.A. Vincenz, Magnetization of the Cambrian Lamotte Formation in Missouri, Geophys. J.R. Astron. Soc. 24 (1971) 175–205.
29 R.A. Fisher, Dispersion on a sphere, Proc. R. Soc. London, Ser. A, 217 (1953) 295–305.
30 C. Bingham, An antipodally symmetric distribution on the sphere, Ann. Statist. 2 (1974) 1201–1225.
31 T.G. Onstott, Application of the Bingham distribution function in paleomagnetic studies, J. Geophys. Res. 85 (1980) 1500–1510.
32 K.V. Mardia, Statistics of Directional Data (Academic Press, London, 1972) 357 pp.